

Molecular Biology & Plant Physiology

Undergraduate Core Notes: Gene Expression, Metabolic Pathways & Cellular Energetics

LEVEL: B.Sc. Biology COURSE: Molecular Genetics & Biochemistry MODULE: 1.0 - Central Dogma

MODULE 1: MOLECULAR GENETICS & CENTRAL DOGMA

The central dogma governs genetic information flow from genomic DNA sequences to functional proteins via messenger RNA intermediaries.

Central Dogma Pathway

DNA Replication (DNA Polymerase) → **Transcription** (RNA Polymerase II) → **Translation** (Ribosome & tRNA machinery).

Key Steps in Gene Expression

- **Transcription:** RNA Polymerase binds to promoter region (TATA box), synthesizing precursor mRNA in 5' to 3' direction.
- **Post-Transcriptional Processing:** 5' Capping, 3' Polyadenylation (Poly-A tail), and Splicing of non-coding introns by spliceosomes.
- **Translation:** Ribosomal sub-units translate codon triplets (AUG start codon) into polypeptide chains via matching anticodons on tRNAs.

MODULE 2: PHOTOSYNTHETIC CARBON FIXATION MECHANISMS

Pathway	Key Enzyme	Anatomical Specialty	Environmental Adaptation
C3 Pathway	RuBisCO	Standard Mesophyll	Temperate / High Moisture
C4 Pathway	PEP Carboxylase	Kranz Anatomy	Hot, Arid, High-light Areas
CAM Pathway	PEP Carboxylase	Succulent Tissue	Extreme Desert / Water Stress

MODULE 3: CELLULAR RESPIRATION & ENERGY YIELD

Cellular respiration converts biochemical energy from nutrients into ATP through sequential oxidation pathways.

Glycolysis (Cytosol)

Anaerobic breakdown of Glucose (6C) into 2 Pyruvate (3C). Yields **2 net ATP** and **2 NADH**.

Krebs Cycle (Mitochondrial Matrix)

Oxidizes Acetyl-CoA, yielding **2 ATP, 6 NADH, and 2 FADH₂** per glucose molecule.

Electron Transport Chain (ETC) & Oxidative Phosphorylation

Complexes I–IV on the inner mitochondrial membrane establish a proton gradient across the intermembrane space, driving ATP synthesis via **ATP Synthase (Chemiosmosis)** for a total yield of ~30–32 ATP per glucose molecule.

Plant Hormones (Phytohormones) Overview

Auxins: Apical dominance & cell elongation | **Cytokinins:** Cell division & delay of senescence | **Gibberellins:** Stem elongation & seed germination | **Abscisic Acid (ABA):** Stress signaling & stomatal closure | **Ethylene:** Fruit ripening & abscission.